

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Functions

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

FUNCTIONS · CH₃ · L₃

ENG. ESLAM AHMED



The Map

This lesson collapses Functions into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Substitute into a function

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find an inverse

TRIGGER *"work out", "find", a missing value*

03 Use composite functions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Solve from function notation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Evaluate fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 31 website questions, 31 checked solutions, 3 local PDFs read.

01 Substitute into a function

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$f(x) = x^2 - 2x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q21

Calculate value

$$f(x) = x^2 - 2x$$

Let

$$y = (x + 2)^2 - 1$$

Finish: $h^{-1}(x) = \sqrt{x+1} - 2$

FORMULA BANK

Substitution for $f(x)$, inverse by swapping x, y , composite means work inside first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

21 The functions f and g are such that $f(x) = x^2 - 2x$ $g(x) = x + 3$ The function h is such that $h(x) = fg(x)$ for $x \geq -2$ Express the inverse function $h^{-1}(x)$ in the form $h^{-1}(x) = \dots$

YOUR SOLUTION

02 Find an inverse

TRIGGER *"work out", "find", a missing value*

BECOMES A Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$f(x) = x^2 - 2x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
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WORKED MODEL

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Let

$$y = (x + 2)^2 - 1$$

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FORMULA BANK

Substitution for $f(x)$, inverse by swapping x, y , composite means work inside first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

21 The functions f and g are such that $f(x) = x^2 - 2x$ $g(x) = x + 3$ The function h is such that $h(x) = fg(x)$ for $x \geq -2$ Express the inverse function $h^{-1}(x)$ in the form $h^{-1}(x) = \dots$

YOUR SOLUTION

03 Use composite functions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$f(x) = x^2 - 2x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q21

Calculate value

$$f(x) = x^2 - 2x$$

Let

$$y = (x + 2)^2 - 1$$

Finish: $h^{-1}(x) = \sqrt{x + 1} - 2$

FORMULA BANK

Substitution for $f(x)$, inverse by swapping x, y , composite means work inside first.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

21 The functions f and g are such that $f(x) = x^2 - 2x$ $g(x) = x + 3$ The function h is such that $h(x) = fg(x)$ for $x \geq -2$ Express the inverse function $h^{-1}(x)$ in the form $h^{-1}(x) = \dots$

YOUR SOLUTION

04 Solve from function notation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$f(x) = x^2 + 6, \quad g(x) = x - 10$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q17

Solve quadratic equation

$$f(x) = x^2 + 6, \quad g(x) = x - 10$$

(a)

$$g(3) = 3 - 10 = -7$$

Finish: $fg(3) = 55$, $x = 5$, excluded value 0, and $h^{-1}(x) = \frac{4}{2-x}$.

FORMULA BANK

Substitution for $f(x)$, inverse by swapping x, y , composite means work inside first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

17 The functions f and g are defined as $f(x) = x^2 + 6$ $g(x) = x - 10$ (a) Find $fg(3)$ (2) (b) Solve the equation $fg(x) = f(x)$ Show clear algebraic working. (3) The function h is defined as $h(x) = 24x -$ (c) State the value of x that cannot be included in the domain of h (1) (d) Express the inverse function h^{-1} in the form $h^{-1}(x) = \dots$ $h^{-1}(x) = (3)$

YOUR SOLUTION

05 Evaluate fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$f(x) = \frac{2x}{x-6}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1H Q14

Evaluate fraction

$$f(x) = \frac{2x}{x-6}$$

(a)

$$f(10) = \frac{2(10)}{10-6} = \frac{20}{4} = 5$$

Finish: $f(10) = 5$, and $f^{-1}(x) = \frac{6x}{x-2}$.

FORMULA BANK

Substitution for $f(x)$, inverse by swapping x, y , composite means work inside first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

14 The function f is defined as $f : x \mapsto 6x - x^2$ (a) Find $f(10)$ (1) (b) Express the inverse function f^{-1} in the form $f^{-1} : x \mapsto \dots$ (3)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Substitute into a function
"work out", "find", a missing value	Find an inverse
"work out", "show", "hence", a familiar exam phrase	Use composite functions
"work out", "show", "hence", a familiar exam phrase	Solve from function notation
"work out", "show", "hence", a familiar exam phrase	Evaluate fraction
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.